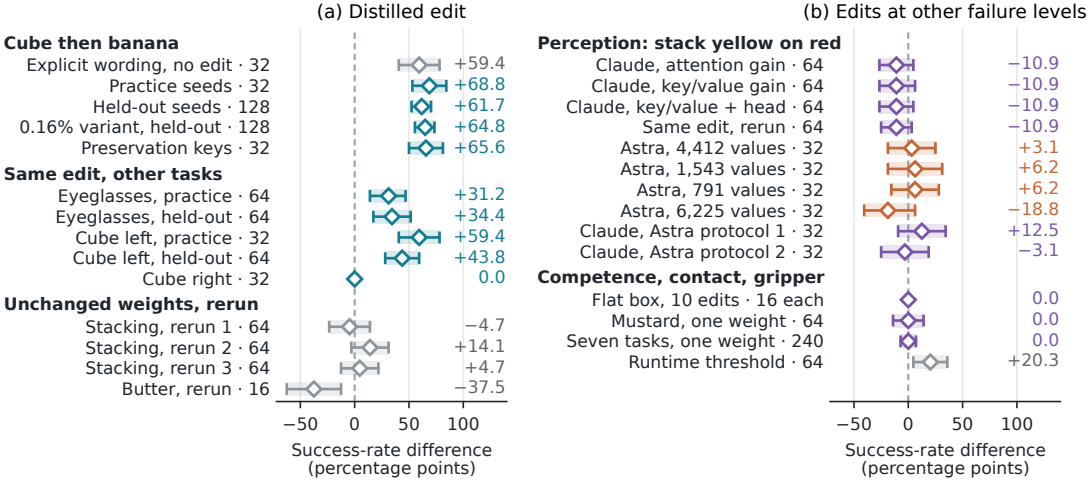
(a) Distilled edit
Cube then banana
Explicit wording, no edit · 32
Practice seeds · 32
Held-out seeds · 128
0.16% variant, held-out · 128
Preservation keys · 32
Same edit, other tasks
Eyeglasses, practice · 64
Eyeglasses, held-out · 64
Cube left, practice · 32
Cube left, held-out · 64
Cube right · 32
Unchanged weights, rerun
Stacking, rerun 1 · 64
Stacking, rerun 2 · 64
Stacking, rerun 3 · 64
Butter, rerun · 16
+59.4
+68.8
+61.7
+64.8
+65.6
+31.2
+34.4
+59.4
+43.8
0.0
−4.7
+14.1
+4.7
−37.5
−50
0
50
100
Success-rate difference
(percentage points)
(b) Edits at other failure levels
Perception: stack yellow on red
Claude, attention gain · 64
Claude, key/value gain · 64
Claude, key/value + head · 64
Same edit, rerun · 64
Astra, 4,412 values · 32
Astra, 1,543 values · 32
Astra, 791 values · 32
Astra, 6,225 values · 32
Claude, Astra protocol 1 · 32
Claude, Astra protocol 2 · 32
Competence, contact, gripper
Flat box, 10 edits · 16 each
Mustard, one weight · 64
Seven tasks, one weight · 240
Runtime threshold · 64
−10.9
−10.9
−10.9
−10.9
+3.1
+6.2
+6.2
−18.8
+12.5
−3.1
0.0
0.0
0.0
+20.3
−50
0
50
100
Success-rate difference
(percentage points)